Learning Task 1 - Music

The purpose of this Learning Task is to let you practise with some of the properties of analog signals, in particular Fourier series and the Fourier transform. We will use the guitar and the drum kit.

Q1 You just bought an acoustic guitar as you would like to learn how to play guitar. The guitar has six strings, tuned to standard EADGBE tuning, see Figure 1 below.



Figure 1: Standard tuning. Source: www.ultimate-guitar.com

You hit the A string only. If we assume the acoustic wave $f_A(t)$ can be described by $f_A(t) = \cos(\omega t)$, what would be $f_A(t)$ with a concrete ω?
H1 The A string corresponds to 440Hz.

Q2 Give the complex Fourier series of $f_A(t)$.
H2 Do you need to calculate anything here?

Q3 Give the Fourier transform of $f_A(t)$ and give an intuitive explanation of your answer.
H3 Table?

Q4 You notice that the Fourier transform of periodic function, here $f_A(t)$, looks similar to the Fourier series. They both have two frequencies at $\omega = \pm 880\pi$. You also see that coefficients are scaled by a factor 2π. Given any Fourier series of a periodic function, predict how the Fourier transform would look like.
H4 Table entry of $e^{j\omega t}$? Also use the fact that the Fourier transform is linear.

Q5 Plucking a single guitar string is only the first stage of learning how to play the guitar. The next level is playing chords. By placing your fingers at certain locations on the neck, you change the length of the strings and hence their frequencies when played[1]. Instead of a single cosine-term, your acoustic signal would be a sum. For example, a D-chord is played on the 4 highest frequency strings (see Figure 2) and is given by

$$f_{Dchord} = \cos(146.8 \cdot 2\pi t) + \cos(220 \cdot 2\pi t) + \cos(293.7 \cdot 2\pi t) + \cos(370 \cdot 2\pi t)$$

For educational purposes (EP) we now consider the acoustic signal

$$f_{EP}(t) = 2\cos(780\pi t) + \cos(1560\pi t)$$

[1] There is very interesting mathematics and physics lurking below this.



Figure 2: D-chord. Source: JustibGuitar.com

Calculate the average power of $f_{EP}(t)$.
H5 Given a complex Fourier series, you can calculate the power of the signal.

Q6 You prefer your sound to be more low frequency, so you want to use a low-pass filter on your guitar. Assume you record your playing using a microphone which converts the acoustic signal to a (time dependent) voltage $V_{in}(t) = x(t)$. This is used as an input voltage to a passive RC filter, producing a (time dependent) output voltage $V_{out}(t) = V_c(t) = y(t)$ as shown in Figure 3; in real life you would of course use a better (active) setup. The frequency response is described

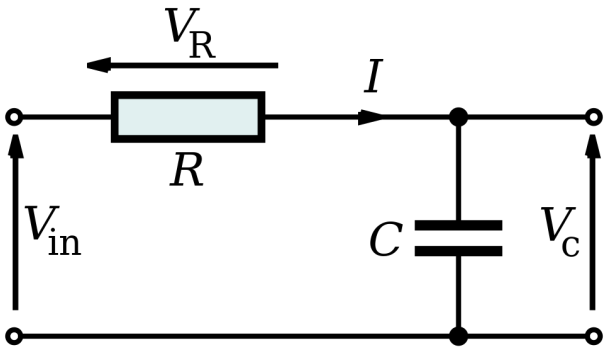


Figure 3: RC circuit. Source: www.wikipedia.org

by $H(\omega)$. This function depends on the values of R and C, and by picking suitable values, you adapt its behaviour. There are various ways to know what $H(\omega)$ is, and it is instructive to see two of those methods. The first approach uses the relationship between any input voltage $V_{in}(t)$ and the output voltage $V_C(t)$. Using some properties of electric circuits and their elements, it can be shown that

$$y'(t) + \frac{1}{RC}y(t) = \frac{1}{RC}x(t).$$

We will not solve this differential equation, but rather use it to illustrate some properties of



Figure 4: Animal abusing the drum kit

the Fourier transform[2]. The filter itself is a linear time invariant (LTI) system[3]. The output $y(t)$ of an LTI system can always by written as a convolution integral between the input $f(t)$ and the impulse response $h(t)$ which is, as its name suggest, the output of the system when provided with the impulse $x(t) = \delta(t)$. Using only the convolution operation, we have

$$y(t) = h(t) \otimes x(t) = x(t) \otimes h(t).$$

Suppose $x(t) \leftrightarrow X(\omega)$, $h(t) \leftrightarrow H(\omega)$, $y(t) \leftrightarrow Y(\omega)$, what is the relationship between $X(\omega)$, $H(\omega)$ and $Y(\omega)$?
H6 Property?

Q7 It can be shown that $h(t) = \frac{1}{RC} e^{-\frac{t}{RC}} u(t)$ for the RC filter. Use the definition of the Fourier transform to derive an expression for $H(\omega)$ and verify your answer by consulting a table.
H7 Look up the definition of the Fourier transform. Why do you think the $u(t)$ and $-$ sign in the exponent are important?

Q8 You can argue that if you have the differential equation $y'(t) + \frac{1}{RC} y(t) = \frac{1}{RC} f(t)$, then by applying the Fourier transform on both sides, you should be able to get $H(\omega)$ as well. Derive $H(\omega)$ through this method.
H8 Properties! There are two routes to the answer, we will show you both of them.

Q9 The -6dB cut-off point of a filter is the frequency ω_c where the power of $H(\omega)$ is half of its maximum power. We want the cut-off point exactly between the two frequencies of $f_{EP}(t)$. Calculate the value of R if you want to use a capacitor of $100nF$ that achieves this.
H9 The instantanuous power of $H(\omega)$ is $|H(\omega)|^2$.

Q10 Plot a graph of $|H(\omega)|$ for $\omega \in [-10000, +10000]$ and visually check the attenuation at the cutoff frequency $\omega = 1170\pi$.
H10 What is an expression for $|H(\omega)|$?

Q11 You are interested in how power calculations in the frequency and time domain can be done. You assume that hitting a drum produces a δ-like acoustic signal $x(t)$. Suppose your filter $x(t)$ with the first order RC circuit with an arbitrary value for RC, yielding output $y(t)$. Calculate the energy of $y(t)$, both in the time and frequency domain.
H11 Convolution, multiplication and another property.

Q12 A realistic linear filter is implemented with physical components and has a real valued impulse response $h(t)$. One can show that if $x(t) = A\cos(\omega_0 t)$ is fed to such a filter with

[2]The Laplace transfrom is a better choice here due to the automatic inclusion of initial conditions.
[3]Albeit under the assumption that the capacitor is fully discharged before the input is supplied

transfer function $H(\omega)$, then the filter output is given by

$$y(t) = A|H(\omega_0)|\cos(\omega_0 t + \arg(H(\omega_0))).$$

The transfer function alters the amplitude and the phase of the input, but leaves the frequency intact. This result is one of the cornerstones of analog signal processing. If you feel up to it, try to prove this. If not, also fine, but try to remember the result and skim over the proof. The proof uses various properties of the Fourier transform.

H12 Go to the frequency domain and return. Use the sifting property of the δ function: $f(\omega)\delta(\omega - \alpha) = f(\alpha)\delta(\omega - \alpha)$ since $\delta(\omega - \alpha) = 0$ for $\omega \neq \alpha$. Finally, write $H(\omega)$ in polar notation $H(\omega) = |H(\omega)|e^{j\arg(H(\omega))}$.